

A Dynamic Model of Humanoid Robots using the Analytical Method

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KEYWORDS: Humanoid robot, Constrained system, Gauss's principle

This paper presents a new approach to analyzing dynamic models of humanoid robots. This new approach is divided into an unconstrained system that applies the principle of Lagrangian dynamics and a constrained system that applies Gauss' principle. Constrained motion is important in the design and analysis of humanoid robots. In this paper, the central issue in the determination of the constrained system is considered to be the determination of this constraint factor. Using the fundamental equation described by the proposed approach, the constraint equation can easily be generated. Based upon the fundamental equation, the dynamic model of the humanoid robot is explained as divided by the sagittal and frontal planes. The suggested approach simplifies designing dynamic models of humanoid robots which will obey the constrained system whether or not they are constrained, holonomic, or open chain.

Manuscript received: August 26, 2008 / Accepted: October 7, 2009

1. Introduction

In general, motion of a humanoid robot in the double support phase can be described as motion under the holonomic constraint imposed by contact between the feet and ground surface in the world frame. The constraining contact is the reason why modeling the double support phase is difficult.²² The main issue in modeling humanoid robot dynamics for walking stability is disturbances depending upon contact condition with the ground. In the humanoid robot research field, most approaches reduce the dynamic model of the robot to that of an inverted pendulum. These approaches are based upon either simple observation or mathematical analysis derived from kinematics. Though these methods are sufficient to model the walking stability of a humanoid robot, they are not enough to solve complex stability problems such as modeling the mobility of human beings.²⁰ An alternate approach is a Lagrange approach that models the constraint dynamics caused by contact with the ground. When Lagrangian dynamics are applied to study the walking stability of humanoid robots, providing stability associated with humanoid posture has been especially successful. This approach is used in planning humanoid robot models which have no knee joints or feet and where motion is constrained to the

sagittal plane.¹⁰ In the future, humanoid robots will become more and more humanlike in their shape and behavior. Miomir Vukobratovic stated that humanoid robots should feature humanlike motion, humanlike intelligence and humanlike interaction.²⁰ Advanced dynamic models for humanoid robot can also be useful tools for dynamic analysis of various human motions. Analysis of humanoid robot systems usually involves developing models representing the robots' physical behavior and predicting the behavior of the dynamic system. It is very difficult to describe accurate dynamic models due to the humanoid robots' nonlinear properties. A fresh way to cope with the complex problems is introduced in this paper. The proposed approach is a simple derivation of analytical dynamics. This approach can be expanded with trajectory tracking control. It provides a unique solution to modeling, predicting, and explaining the behavior of complex humanoid robot systems without relying upon Lagrangian multipliers, which are problem-specific and often difficult to obtain for systems with a large number of degrees of freedom and many non-integrable constraints.² This approach of dynamic modeling for humanoid robot is based upon the theory by Udwadia and Kalaba,^{1,5-8} who provide a new and different perspective on the constrained motion of multi-body systems.² The theory is applied to

constraints the multi-body system may be subjected to where the constraint force is an ideal constraint, which does not work under virtual displacement. In most real situations where the constrained motion involves sliding, the constraint is said to be non-ideal. The assumption that total work done by constraint forces is equal to zero is no longer true if sliding friction forces are present.⁵ However, the range of non-ideal constraints is beyond the scope of this paper, which only handles ideal constraints without friction. In this paper, how to obtain the fundamental equation of motion for a constrained system is presented first. Unconstrained and constrained systems are considered separately. In unconstrained systems, the number of coordinates required to specify the configuration of the system is equal to its degree of freedom. By impressing a force on the system, the total equation of motion can be expressed by the fundamental equation using generalized coordinates. This is realized by minimization using Gauss' principle. Secondly, the fundamental equation is applied to describe the dynamic motion of a humanoid robot where the system is assumed as frictionless. The constraint fundamental equation shows that the model is to be split into two parts by the sagittal plane and the frontal plane. Lastly, torque value of each pitch joint is examined using dynamic equation in the sagittal plane derived from the fundamental equation, with the results compared to a commercial tool for feasibility.

2. Fundamental Equation

The dynamic system is considered as a whole and scalar quantities such as energy and work are used. Constraint properties in analytical mechanics are treated differently than in Newtonian mechanics. Constraint forces that do not work do not appear in the formulation and are accounted for by appropriately selecting the variables used to describe the motion under study. Analytical techniques use scalar functions like work and energy in the formulation rather than vector quantities. In dynamic systems, constraints are usually uncoupled as a result of contact between two or more bodies. Constraints restrict the motion of the bodies on which they act; a body is associated with a constraint through a constraint equation which describes the geometry and kinematics of the contact. The constraint force is the same as the contact force. An unconstrained dynamical system, or one subjected to a holonomic constraint that is not an explicit function of time $f_j(q_1 \cdots q_n) = 0$ is called a scleronomic system. If the holonomic constraint is an explicit function of time, the system is rheonomic. When the constraint is nonholonomic, there is no integrating factor to permit expression. None of the preceding results obtained regarding the work done by the constraint force are valid for nonholonomic constraints. The constraint force associated with a nonholonomic constraint cannot be expressed as a force normal to a surface, since the nonholonomic constraint does not define a surface. A constraint of the form $f_j(q_1 \cdots q_n, t) \geq 0$, or an inequality constraint, is nonholonomic because it cannot be reduced to the form $f_j(q_1 \cdots q_n) = 0$; such constraints require different treatment than equality constraints. There are also constraints that are only valid

when the body is in a certain position or during certain intervals of the motion being studied. Such constraints are classified as equality constraints. These are found in problems involving contact. The central issue here is the determination of the time evolution of the constrained system, with the constraint as force vector F^c . This vector can be determined in accordance with the accepted principle of analytical dynamics, $M\ddot{X} = F(t) + F^c(t)$. Though the form of these equations of motion may be known and look like dynamic equations, they are unknown in content because the constraint force vector $F^c(t)$ is unknown. This constraint force must be such that it acts in combination with the given impressed force $F(t)$. Two different types of constraints have been noted to appear to be of importance in practical situations that arise in mechanics: integrable and nonintegrable. As nonintegrable constraints are more difficult to handle, they have engendered special methods for their analysis. However, this paper shows that both constraints can be handled with equal ease by the methods developed in this study, contrary to accepted practice. No new concept is required to handle nonintegrable as opposed to integrable constraints. The equation of the motion of the constrained system described at each instant of time is expressed using generalized coordinates. Consequently, Eq. (1) is the fundamental equation under constrained systems. The procedure for developing the fundamental equation follows below.

$$M(q, t)\ddot{q}(t) = F(q, \dot{q}, t) + M^{\frac{1}{2}}(AM^{-\frac{1}{2}})^+(b - Aa) \quad (1)$$

As mentioned earlier, the central issue in the research of constrained motion is the determination of the acceleration $\ddot{q}(t)$ of the constrained system at that instant of time t . Its acceleration $\ddot{q}$ at time t will differ from what it would have been had there been no constraint on it. In fact, its acceleration in the absence of constraints would be exactly $a(t)$. $a(t)$ is determined using $q(t)$ and $\dot{q}(t)$, which may be required to determine F at time t . However, the acceleration $a(t)$ as obtained will not satisfy the constraint equation in general. We can denote this difference vector at each instant of time t using Eq. (2).⁸

$$e(t) = (b - Aa), \quad F^c = M^{\frac{1}{2}}(AM^{-\frac{1}{2}})^+e = Ke \quad (2)$$

If A is a square nonsingular matrix, then the usual inverse of A , matrix A^{-1} , satisfies all four of the MP conditions (Appendix A). Gauss' principle gives a clear description of the general nature of constrained motion in terms of the minimization of a function of the acceleration of a system.^{6,9}

$$G(q) = [\ddot{q} - a(t)]^T M(q, t) [\ddot{q} - a(t)] \quad (3)$$

The fundamental equation is proved in generalized coordinates. The equation for the unconstrained system is known to always be defined in the form of an equation such as below.

$$M(q, t)\ddot{q} = F(q, \dot{q}, t) \quad (4)$$

The set of w constraints is assumed to be expressed in the form $\varphi_i(q, \dot{q}, t) = 0$. These w constraint equations are differentiated, and then the equation of constraint is defined in this form.

$$A(q, \dot{q}, t)\ddot{q} = b(q, \dot{q}, t) \quad (5)$$

Here, the vector q is s vector and A is w by s , where $i=1\cdots w$ constraint number. The matrix A may be a function of q , $\dot{q}$, and t . If A is a positive definite matrix, we can define it as

$$\begin{aligned} A^{\frac{1}{2}} &= W \Lambda^{\frac{1}{2}} W^T, & A^{-\frac{1}{2}} &= W A^{-\frac{1}{2}} W^T \\ AW &= W \Lambda, & W^T W &= I \end{aligned}$$

Where W is an orthogonal matrix and Λ is a diagonal matrix. In the absence of the constraint Eq. (5) the acceleration $\ddot{q}(t) = a(t)$ of the unconstrained system is given by using Eq. (4).

$$a(t) = M^{-1}(q, t) F(q, \dot{q}, t) \quad (6)$$

In general, the matrix A has rank $r \leq w$ where r is the number of row vectors. Additionally, w constraint equations need not be independent. The acceleration $\ddot{q}(t)$ of the constraint equation at each instant of time t must satisfy Eq. (5). For clarity, the arguments of various matrices and vectors have been suppressed, and matrix M is assumed to be positive definite. If vector $\ddot{r}$ is then defined,

$$\ddot{r}(t) = M^{\frac{1}{2}}(q, t) \ddot{q}(t) \quad (7)$$

Eq. (7) can be substituted into the constraint Eq. (5). The acceleration of the constrained system then satisfies Eq. (8).

$$A(q, t) M^{-\frac{1}{2}}(q, t) \ddot{r}(t) = b(q, \dot{q}, t) \quad (8)$$

$B = AM^{-1/2}$ is defined as a constraint matrix, where B is the w by s matrix.

$$B(q, t) \ddot{r}(t) = b(q, \dot{q}, t) \quad (9)$$

Eq. (10) is the general solution to the consistent equation, $AX = b$, where the matrix A is m by n and h is any n by 1 vector (See Appendix B).

$$X = A^+ b + (I - A^+ A) h \quad (10)$$

In the same sense, the general solution to Eq. (9) is simple.

$$\ddot{r}(t) = B^+ b + (I - B^+ B) h \quad (11)$$

Where B^+ is an s by w matrix and the MP-inverse of the matrix B , and h is an arbitrary n vector. This is then redefined.

$$\ddot{r}(t) = B^+ b + Rh \quad \text{where, } R = (I - B^+ B) \quad (12)$$

The vector Rh needs to be defined in accordance with Gauss' principle in analytical mechanics. The acceleration of the constrained system must satisfy the constraint equation at time t . Among the accelerations, Gauss' principle minimizes the constraint which is a deviation from the acceleration of the constrained system, such as $\Delta \ddot{q}(t) = \ddot{q}(t) - a(t)$. The quantity $G(\ddot{q})$ is the square of the normalized length of the vector $\Delta \ddot{q}(t)$, normalized with respect to matrix M . If minimized, $G(\ddot{q})$ is given by this equation.

$$\begin{aligned} G(\ddot{q}) &= (\ddot{q} - a)^T M (\ddot{q} - a) \\ &= (M^{\frac{1}{2}} \ddot{q} - M^{\frac{1}{2}} a)^T (M^{\frac{1}{2}} \ddot{q} - M^{\frac{1}{2}} a) \\ &= (\ddot{r} - M^{\frac{1}{2}} a)^T (\ddot{r} - M^{\frac{1}{2}} a) \end{aligned} \quad (13)$$

Substituting Eq. (11) into (13), vector h needs to be found such that the next equation is minimized

$$G(h) = (Rh - (M^{\frac{1}{2}} a - B^+ b))^T (Rh - (M^{\frac{1}{2}} a - B^+ b)) \quad (14)$$

Hence, if the matrix A is m by n and b is m vector, an n vector X is minimized by

$$K(x) = \|AX - b\|^2 = (AX - b)^T (AX - b) \quad (15)$$

given that $X = A^+ b + (I - A^+ A)h$.

Therefore, Eq. (16) is obtained to minimize $G(h)$,

$$h = R^+ (M^{\frac{1}{2}} a - B^+ b) + (I - R^+ R) K \quad (16)$$

where K is an arbitrary n vector. Since $R = (I - B^+ B)$ is defined, R is materialized through $RR = R, R^+ = R$ to get

$$h = R^+ (M^{\frac{1}{2}} a - B^+ b) + (I - R) K \quad (17)$$

Both sides are multiplied by R .

$$\begin{aligned} Rh &= RR^+ (M^{\frac{1}{2}} a - B^+ b) + R(I - R) K \\ &= RR(M^{\frac{1}{2}} a - B^+ b) + (R - RR) K \\ &= R(M^{\frac{1}{2}} a - B^+ b) \end{aligned} \quad (18)$$

Eq. (18) is substituted into Eq. (11).

$$\begin{aligned} \ddot{r}(t) &= B^+ b + R(M^{\frac{1}{2}} a - B^+ b) \\ &= B^+ b + (I - B^+ B)(M^{\frac{1}{2}} a - B^+ b) \\ &= M^{\frac{1}{2}} a + B^+ (b - BM^{\frac{1}{2}} a) \\ &= M^{\frac{1}{2}} a + (AM^{-\frac{1}{2}})^+ (b - Aa) \end{aligned} \quad (19)$$

Where $B = AM^{-1/2}$ is the constraint matrix,

$$\begin{aligned} \ddot{r}(t) &= M^{\frac{1}{2}}(q, t) \ddot{q}(t) \\ \ddot{q}(t) &= a + M^{-\frac{1}{2}}(AM^{-\frac{1}{2}})^+ (b - Aa) \end{aligned} \quad (20)$$

This is the total acceleration of constrained system. Both sides of Eq. (20) can be multiplied by M to get this.

$$\begin{aligned} M\ddot{q}(t) &= Ma + M^{\frac{1}{2}}(AM^{-\frac{1}{2}})^+ (b - Aa) \\ &= F(t) + F^c(t) \end{aligned} \quad (21)$$

The unconstrained equation is $Ma(t) = F(t)$ and the constrained equation is $F^c(t) = M^{1/2}(AM^{-1/2})^+(b - Aa)$. The force $F(t)$ is the given force which is impressed on the system. The vector $F^c(t)$ satisfies the constraint Eq. (5). The equation of motion is rewritten using the vector λ , known as the Lagrange multiplier in classical dynamics. λ is then assumed to be an arbitrary w -vector.

$$\begin{aligned} F^c(t) &= M^{\frac{1}{2}}(AM^{-\frac{1}{2}})^+ (b - Aa) \\ &= A^T (AM^{-1} A^T)^+ (b - Aa) \\ &= A^T (AM^{-1} A^T)^{-1} (b - Aa) \\ &= A^T \lambda \end{aligned} \quad (22)$$

When the rank of A is w , $(AM^{-1/2})^+ = (AM^{-1/2})^T$ are identical. It should be pointed out that the fundamental equation, Eq. (21), can be used with equal ease for both nonholonomic and holonomic constraints whether the constraint equation $\varphi(q, \dot{q}, t) = 0$ is dependent or not. However, when the equation of motion, Eq. (22), is rewritten classically using Lagrange multipliers, the Lagrange multipliers at each instant of time are non-unique, if constraint equations are not independent or when dealing with complex systems as is the case in most practical situations, because arbitrary quantities are included. In fact, acceleration in the absence of constraints would be exactly $a(t)$; $a(t)$ is determined by $q(t)$ and $\dot{q}(t)$, but the acceleration $a(t)$ obtained will not satisfy the constraint Eq. (5) when $\ddot{q}(t)$ is replaced by $a(t)$. As stated above, $\ddot{q}(t) - a(t)$ is simply the deviation of the acceleration between unconstrained and constrained systems. This is shown in $\Delta\ddot{q}(t) = \ddot{q}(t) - a(t)$. The quantity G is the square of the normalized length of the vector $\Delta\ddot{q}(t)$, normalized with respect to the matrix M . Clearly, no constraint for the minimum of $G(\ddot{q})$ is achieved when $\ddot{q}(t) = a(t)$ and $\Delta\ddot{q}(t) = 0$. The unique solution for the acceleration $\ddot{q}(t)$ to Gauss' minimization is understood to be described by Eq. (21). For that reason, the fundamental equation is appropriately classified as developed under Gauss' principle. The fundamental equation satisfies any system—whether it is constrained or unconstrained, holonomic or nonholonomic, dependent or independent—in contrast to using Lagrange multipliers.

3. Dynamic Model for Humanoid Robots in the Sagittal Plane

In this paper, the motion of the humanoid robot is described in the dynamic model as consisting of two planes, divided by the sagittal and frontal planes; this planar model can be compared to a chain of n rigid links. In the case of the sagittal plane, the motion is defined with 10 variables: the position (X_B, Z_B) of the base; the

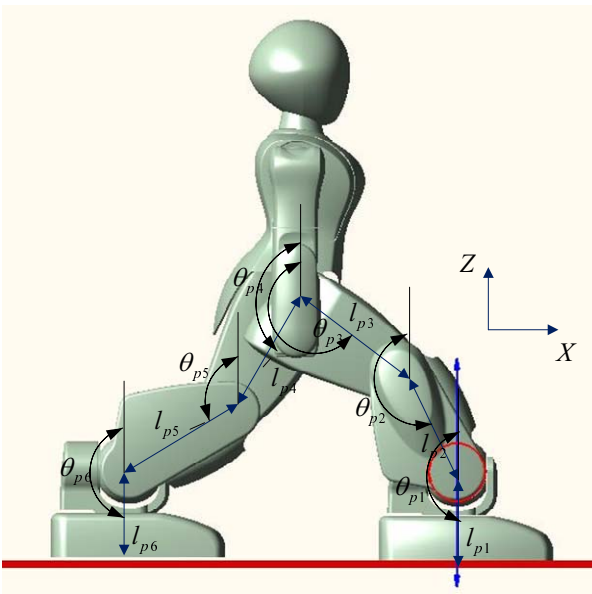


Fig. 1 Sagittal plane

position (X_o, Z_o) of foot contact with respect to the base coordinates; and angles θ_{p1} , θ_{p2} and θ_{p3} for the right leg and θ_{p4} , θ_{p5} and θ_{p6} for the left leg. The support leg is assumed not to slide and be fixed at the contact point, and the angle of trunk is assumed not to rotate during motion. The fundamental equation for humanoid robot is shown here to be very useful in describing complex dynamic systems. In the first part, the unconstrained system described by n generalized coordinates q_i is presented, and can be obtained using the Lagrange equation of motion. Since a double support phase is being considered, the kinematical model of unconstrained system is given by these two equations.

$$\left. \begin{aligned} X_{c1} &= l_{c1} \sin \theta_{p1} \\ X_{c2} &= l_1 \sin \theta_{p1} + l_{c2} \sin \theta_{p2} \\ X_{c3} &= l_1 \sin \theta_{p1} + l_2 \sin \theta_{p2} + l_{c3} \sin \theta_{p3} \\ X_{c4} &= X_B - l_{c4} \sin \theta_{p4} \\ X_{c5} &= X_B - l_4 \sin \theta_{p4} + l_{c5} \sin \theta_{p5} \\ X_{c6} &= X_B - l_4 \sin \theta_{p4} + l_5 \sin \theta_{p5} + l_{c6} \sin \theta_{p6} \end{aligned} \right\} \quad (23)$$

$$\left. \begin{aligned} Z_{c1} &= -l_{c1} \cos \theta_{p1} \\ Z_{c2} &= -l_1 \cos \theta_{p1} - l_{c2} \cos \theta_{p2} \\ Z_{c3} &= -l_1 \cos \theta_{p1} - l_2 \cos \theta_{p2} - l_{c3} \cos \theta_{p3} \\ Z_{c4} &= Z_B + l_{c4} \cos \theta_{p4} \\ Z_{c5} &= Z_B + l_4 \cos \theta_{p4} + l_{c5} \cos \theta_{p5} \\ Z_{c6} &= Z_B + l_4 \cos \theta_{p4} + l_5 \cos \theta_{p5} + l_{c6} \cos \theta_{p6} \end{aligned} \right\} \quad (24)$$

Where $X_B = l_1 \sin \theta_{p1} + l_2 \sin \theta_{p2} + l_4 \sin \theta_{p4}$ and $Z_B = -l_1 \cos \theta_{p1} - l_2 \cos \theta_{p2} - l_3 \cos \theta_{p3}$.

The Lagrange formulation for unconstrained equations is given below.

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}_{pi}} \right) - \frac{\partial T}{\partial \theta_{pi}} + \frac{\partial V}{\partial \theta_{pi}} = \tau_{pi} \quad (25)$$

T is kinetic energy with rotation and V is potential energy. $L = T - V$ is the Lagrangian of the system, and τ_{pi} is the unconstrained force.

$$T = \frac{1}{2} \sum_{i=1}^6 (m_i v_i^2 + I_i \dot{\theta}_{pi}^2), \quad i = 1 \cdots 6 \quad (26)$$

$$\left. \begin{aligned} V_1 &= -m_1 g (l_1 \cos \theta_{p1}) \\ V_2 &= -m_2 g (l_1 \cos \theta_{p1} + l_{c2} \cos \theta_{p2}) \\ V_3 &= -m_3 g (l_1 \cos \theta_{p1} + l_2 \cos \theta_{p2} + l_{c3} \cos \theta_{p3}) \\ V_4 &= -m_4 g (l_1 \cos \theta_{p1} + l_2 \cos \theta_{p2} + l_3 \cos \theta_{p3} - l_{c4} \cos \theta_{p4}) \\ V_5 &= -m_5 g (l_1 \cos \theta_{p1} + l_2 \cos \theta_{p2} + l_3 \cos \theta_{p3} \\ &\quad - l_4 \cos \theta_{p4} - l_{c5} \cos \theta_{p5}) \\ V_6 &= -m_6 g (l_1 \cos \theta_{p1} + l_2 \cos \theta_{p2} + l_3 \cos \theta_{p3} \\ &\quad - l_4 \cos \theta_{p4} - l_5 \cos \theta_{p5} - l_{c6} \cos \theta_{p6}) \\ V &= \sum_{j=1}^6 V_j, j = 1 \cdots 6 \end{aligned} \right\} \quad (27)$$

This is substituted into the unconstrained form given here.

$$\begin{pmatrix} m_{11} & \dots & m_{16} \\ \vdots & \ddots & \vdots \\ m_{61} & \dots & m_{66} \end{pmatrix} \begin{pmatrix} \ddot{\theta}_{p1} \\ \vdots \\ \ddot{\theta}_{p6} \end{pmatrix} + \begin{pmatrix} N_{p1} \\ \vdots \\ N_{p6} \end{pmatrix} = \begin{pmatrix} \tau_{p1} \\ \vdots \\ \tau_{p6} \end{pmatrix} \quad (28)$$

The equation of motion using classical dynamics about unconstrained is obtained. Eq. (28) is rewritten to be expressed as in the form $M\ddot{q} = F - N$. For the second task, the constrained system described by Gauss' minimization is presented. The basic form of constraint equations is $\varphi(q, \dot{q}, t) = 0$, $i = 1 \dots w$. As stated above, these constraint equations are differentiated. The result is $A(q, t)\ddot{q} = b(q, \dot{q}, t)$. The equation of motion of the constrained system described at each instant of time is expressed in generalized coordinates, and Eq. (29) is obtained.

$$-\sum_{i=1}^6 (l_{ci} \sin \theta_{pi}) = \sum C_i \quad (29)$$

Where C_i is a constant value about $i = 1 \dots 6$. Eq. (29) is then differentiated twice.

$$\sum_{i=1}^6 (l_{ci} \sin \theta_{pi} \dot{\theta}_{pi}^2 - l_{ci} \cos \theta_{pi} \ddot{\theta}_{pi}) = 0 \quad (30)$$

This is substituted into the constraint form given here.

$$\begin{bmatrix} l_{c1} \cos \theta_{p1} & \dots & l_{c6} \cos \theta_{p6} \\ \vdots & & \vdots \\ l_{c1} \sin \theta_{p1}^2 & \dots & l_{c6} \sin \theta_{p6}^2 \end{bmatrix} \begin{pmatrix} \ddot{\theta}_{p1} \\ \vdots \\ \ddot{\theta}_{p6} \end{pmatrix} = \begin{bmatrix} \ddot{\theta}_{p1} \\ \vdots \\ \ddot{\theta}_{p6} \end{bmatrix} \quad (31)$$

The fundamental equation of motion, Eq. (21), expressed from $M_p \ddot{q}_p(t) = F_p(t) + F_p^c(t)$ is recalled here.

$$M_p \ddot{q}_p(t) = M_p a_p + M_p^{-\frac{1}{2}} (A_p M_p^{-\frac{1}{2}})^+ (b_p - A_p a_p) \quad (32)$$

Where,

$$F_p(t) = M_p a_p$$

$$F_p^c(t) = M_p^{-\frac{1}{2}} (A_p M_p^{-\frac{1}{2}})^+ (b_p - A_p a_p)$$

Finally, the fundamental equation of motion of constrained systems for humanoid robots in the sagittal plane is described here.

$$M_p \ddot{q}_p(t) = \tau_p - N_p + M_p^{-\frac{1}{2}} (A_p M_p^{-\frac{1}{2}})^+ (b_p - A_p M_p^{-1} F_p) \quad (33)$$

M_p is a w by s vector, $\ddot{q}_p$ is an s -vector, A_p is a w vector, and F_p is $\tau_p - N_p$, where $w = 1 \dots 6$, $s = 1 \dots 6$.

4. Dynamic Model for Humanoid Robots in the Frontal Plane

In the case of the frontal plane, the motion is defined with 8 variables: the position (X_B, Z_B) of the base; the position (X_o, Z_o) of foot contact with respect to base coordinates; and angles θ_{r1} ,

θ_{r2} for the right leg and θ_{r3} , θ_{r4} for the left. Since a single support phase is being considered, the kinematical model of unconstrained system in the frontal plane is given by these equations.

$$\left. \begin{aligned} Y_{c1} &= l'_{c1} \sin \theta_{r1} \\ Y_{c2} &= l'_1 \sin \theta_{r1} + l'_{c2} \sin \theta_{r2} \\ Y_{c3} &= Y_{B'} - l'_{c3} \sin \theta_{r3} \\ Y_{c4} &= Y_{B'} - l'_{c3} \sin \theta_{r3} - l'_{c4} \sin \theta_{r4} \end{aligned} \right\} \quad (34)$$

$$\left. \begin{aligned} Z_{c1} &= -l'_{c1} \cos \theta_{r1} \\ Z_{c2} &= -l'_1 \cos \theta_{r1} - l'_{c2} \cos \theta_{r2} \\ Z_{c3} &= Z_{B'} + l'_{c3} \cos \theta_{r3} \\ Z_{c4} &= Z_{B'} + l'_{c3} \cos \theta_{r3} + l'_{c4} \cos \theta_{r4} \end{aligned} \right\} \quad (35)$$

$$\left. \begin{aligned} Y_{B'} &= l'_1 \sin \theta_{r1} + l'_2 \sin \theta_{r2} \\ Z_{B'} &= -l'_1 \cos \theta_{r1} - l'_2 \cos \theta_{r2} \\ l'_1 &= l_1 \\ l'_4 &= l_4 \\ l'_2 &= l_2^2 + l_3^2 - 2l_2 l_3 \cos(\theta_{p2} - \theta_{p3} + \pi) \\ l'_3 &= l_4^2 + l_5^2 - 2l_4 l_5 \cos(\theta_{p4} - \theta_{p5} + \pi) \end{aligned} \right\} \quad (36)$$

Similar to the previous section, the Lagrange formulation for unconstrained equation in the frontal plane is known and given in Eq. (37).

$$\frac{d}{dt} \left(\frac{\partial T'}{\partial \dot{\theta}_{ri}} \right) - \frac{\partial T'}{\partial \theta_{ri}} + \frac{\partial V'}{\partial \theta_{ri}} = \tau_{ri} \quad (37)$$

Where $i = 1 \dots 4$. $\dot{L} = T' - V'$ is substituted into Eq. (25) by the same formulation.

$$\begin{aligned} T' &= \frac{1}{2} \sum_{i=1}^4 (m_i v_i^2 + I_i \dot{\theta}_{ri}^2) \quad i = 1 \dots 6 \\ V' &= -m'_1 g l'_{c1} \\ V'_2 &= -m'_2 g (l'_1 \cos \theta_{r1} + l'_{c2} \cos \theta_{r2}) \\ V'_3 &= -m'_3 g (l'_1 \cos \theta_{r1} + l'_2 \cos \theta_{r2} - l'_{c3} \cos \theta_{r3}) \\ V'_4 &= -m'_4 g (l'_1 \cos \theta_{r1} + l'_2 \cos \theta_{r2} - l'_3 \cos \theta_{r3} - l'_{c4} \cos \theta_{r4}) \\ V' &= \sum_{j=1}^4 V'_j \quad j = 1 \dots 4 \end{aligned} \quad (38)$$

This is substituted into its unconstrained form in the frontal plane, given here.

$$\begin{pmatrix} m'_{11} & \dots & m'_{14} \\ \vdots & \ddots & \vdots \\ m'_{41} & \dots & m'_{44} \end{pmatrix} \begin{pmatrix} \ddot{\theta}_{r1} \\ \vdots \\ \ddot{\theta}_{r4} \end{pmatrix} + \begin{pmatrix} N_{r1} \\ \vdots \\ N_{r4} \end{pmatrix} = \begin{pmatrix} \tau_{r1} \\ \vdots \\ \tau_{r4} \end{pmatrix} \quad (39)$$

The equation of motion of the constrained system described at each instant of time is expressed in generalized coordinates, and then the constraint equation is differentiated twice.

$$\sum_{i=1}^4 (l'_{ci} \sin \theta_{ri} \dot{\theta}_{ri}^2 - l'_{ci} \cos \theta_{ri} \ddot{\theta}_{ri}) = 0, \quad i = 1 \dots 4 \quad (40)$$

The constrained system described by the Gauss's minimization is

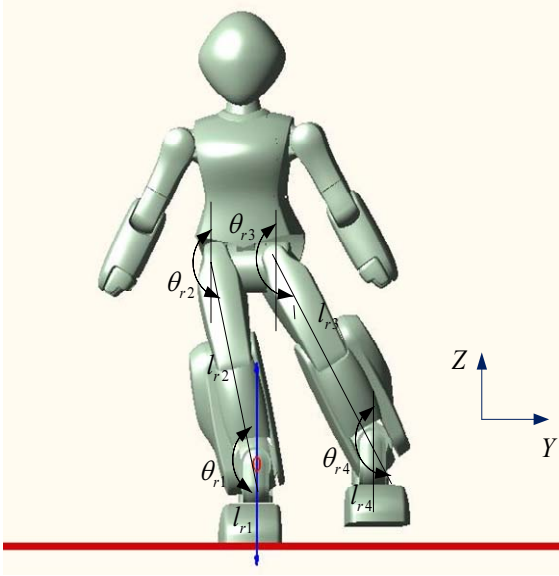


Fig. 2 Frontal plane

presented as stated above. The basic form of constraint equations is $\varphi(q, \dot{q}, t) = 0$, $i = 1 \dots w$.

$$\underbrace{\begin{bmatrix} l'_{c1} \cos \theta_{r1} & \dots & l'_{c4} \cos \theta_{r4} \end{bmatrix}}_{A_r} \begin{bmatrix} \ddot{\theta}_{r1} \\ \vdots \\ \ddot{\theta}_{r4} \end{bmatrix} = \underbrace{\begin{bmatrix} l'_{c1} \sin \theta_{r1}^2 & \dots & l'_{c4} \sin \theta_{r4}^2 \end{bmatrix}}_{b_r} \ddot{q}_r \quad (41)$$

The fundamental equation of motion, Eq. (21), expressed from $M_r \ddot{q}_r(t) = F_r(t) + F_r^c(t)$ is recalled here.

$$M_r \ddot{q}_r(t) = M_r a_r + M_r^{\frac{1}{2}} (A_r M_r^{-\frac{1}{2}})^+ (b_r - A_r a_r) \quad (42)$$

The fundamental equation of motion of constrained system for humanoid robot in the frontal plane is described here.

$$M_r \ddot{q}_r(t) = \tau_r - N_r + M_r^{\frac{1}{2}} (A_r M_r^{-\frac{1}{2}})^+ (b_r - A_r M_r^{-1} F_r) \quad (43)$$

In conclusion, the total fundamental equation of constrained humanoid robot is expressed by the constraint equation in the sagittal plane (Eq. (33)) and the constraint equation in the frontal plane (Eq. (42)). The fundamental equation of motion of the humanoid robot is summarized below.

$$\begin{bmatrix} m_{11} & \dots & m_{16} \\ \vdots & \ddots & \vdots \\ m_{61} & \dots & m_{66} \\ m'_{11} & \dots & m'_{14} \\ \vdots & \ddots & \vdots \\ m'_{41} & \dots & m'_{44} \end{bmatrix} \begin{bmatrix} \ddot{\theta}_{p1} \\ \vdots \\ \ddot{\theta}_{p6} \\ \ddot{\theta}_{r1} \\ \vdots \\ \ddot{\theta}_{r4} \end{bmatrix} = \begin{bmatrix} F_{p1} + F_{p1}^c \\ \vdots \\ F_{p6} + F_{p6}^c \\ F_{r1} + F_{r1}^c \\ \vdots \\ F_{r4} + F_{r4}^c \end{bmatrix} \quad (44)$$

Where $\sum F_{pi} = \sum_{i=1}^6 (\tau_{pi} - N_{pi})$, $\sum F_{ri} = \sum_{i=1}^4 (\tau_{ri} - N_{ri})$, $m_{11} \dots m_{66}$ are inertia matrix components in sagittal plane and $m'_{11} \dots m'_{44}$ are inertia matrix components in frontal plane.

5. Evaluation

In this section, it is necessary to evaluate the accuracy of dynamic model obtained by fundamental equation. To characterize the walking pattern, the humanoid robot is modeled as a point mass,

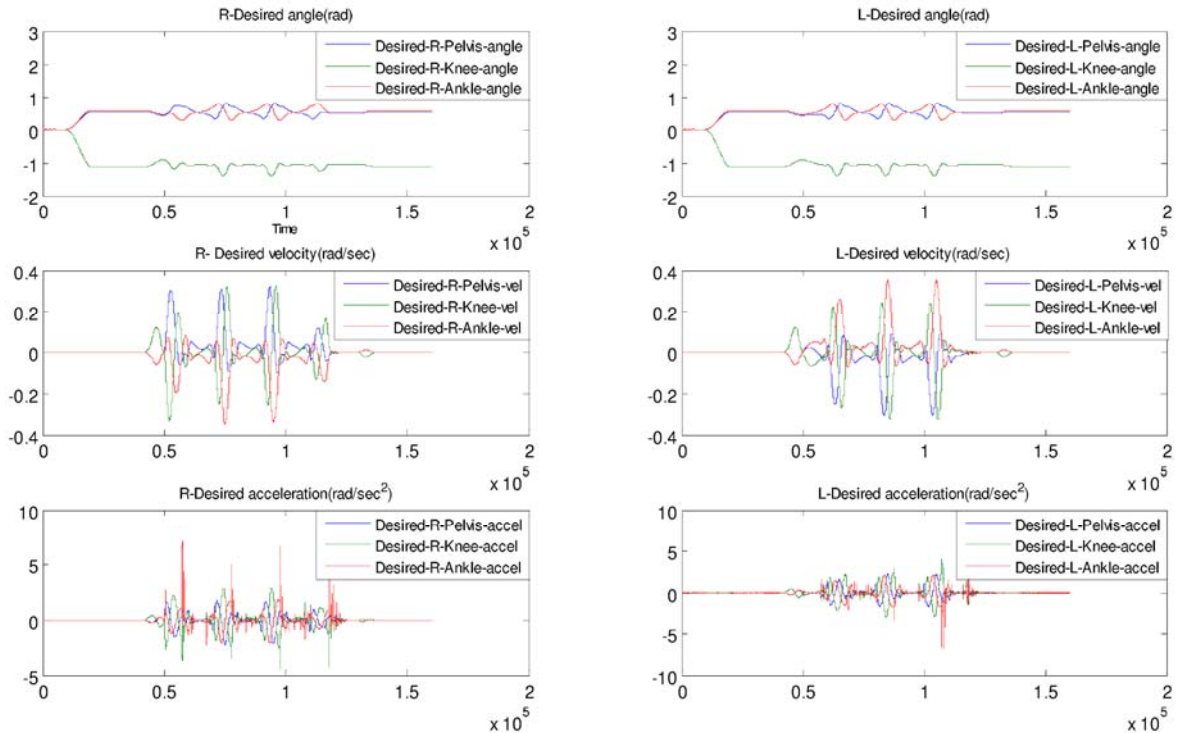


Fig. 3 Desired Pitch Joint Angles, velocities and accelerations

Table 1 dynamic parameters

mass	value	link length	value	Inertia	value
M_{p1}	1.2	l_1	0.0125	I_{p1}	0.0026
M_{p2}	1.8	l_2	0.155	I_{p2}	0.0102
M_{p3}	2.0	l_3	0.1625	I_{p3}	0.0094
M_{p4}	2.0	l_4	0.0625	I_{p4}	0.0094
M_{p5}	1.8	l_5	0.0155	I_{p5}	0.0102
M_{p6}	1.2	l_6	0.0125	I_{p6}	0.0026
M_B	3.1	l_B	0.21	I_B	0.0295

called LIPM (linear inverted pendulum model) which treats the behavior of its COM (the center of mass) to generate walking pattern trajectory. In the LIPM, a point mass connected to a swing leg is maintained at a constant height and is supported by a support foot. The LIPM is linearized about the vertical and constrained to a horizontal plane. The X -direction trajectory of COM is given by

$$x_G(t) = \left(\frac{f_r}{\omega^2} + x_0\right) \cosh(\omega t) + \frac{\dot{x}_0}{\omega} \sinh(\omega t) - \frac{f_r}{\omega^2} \quad (45)$$

Where $\omega = \sqrt{\frac{g}{H_z}}$ and $f_r = -\frac{g}{H_z} p_x$.

Here, the initial conditions of COM position and velocity are $x(0) = x_0$, and $\dot{x}(0) = \dot{x}_0$ respectively. H_z is the constant height of the hip link from the ground, g is gravitational acceleration, and p_x is a ZMP about x -direction. The Y -component of the COM can be found in a similar manner. The walking pattern of the swing leg is generated by the cubic polynomial. When the Cartesian position, velocity and acceleration values are computed, the joint space values such as rotational positions, velocities, and accelerations can

be calculated by using the following inverse kinematic relations.

$$\begin{aligned} Q &= T^{-1}P \\ \dot{Q} &= J_p^{-1}\dot{P} \\ \ddot{Q} &= J_p^{-1}\ddot{P} - J_p^{-1}\dot{J}_p J_p^{-1}\dot{P} \end{aligned} \quad (46)$$

Here, $Q^T = [\theta_1 \cdots \theta_6]$ is the link rotation vector, $P^T = [x, y, z]$ is the Cartesian coordinate vector, T^{-1} is the inverse of the transformation matrix, J_p^{-1} is the inverse of the Jacobian matrix and $\dot{J}_p$ is the time derivative of the Jacobian matrix. Using the rotational positions, velocities, accelerations, it is possible to compute joint torques by the dynamic equation. The verification can be evaluated by RMS (root mean square) torque difference between the dynamic model in the commercial software tool (Recurdyn 6.3) and the fundamental equation with reference pitch joint values (Fig. 3). The dynamic parameters (Table 1) are applied to both dynamic models. In the fig. 4, it is shown that RMS torque differences of dynamic model with constraint (red line) are smaller than those of dynamic model without constraint (blue line). Hence, the robot dynamic model with constraint is more accurate than that of no constraint. Suppose that the sampling time is 10msec, the maximum variation about RMS torque difference between with-constraint and without- constraint is 0.0987Nm in the right ankle joint (Table 2). As a result, the dynamic model of the humanoid robot with constraint is shown to be gotten approximately similar to commercial dynamic model. When the sampling time is faster, we estimate that RMS torque difference between with-constraint and without- constraint also increases. In addition, considering other constraints such as link frictions or between the feet and ground.etc, the equation of motion of humanoid robots with those constraints will be closed to real system.

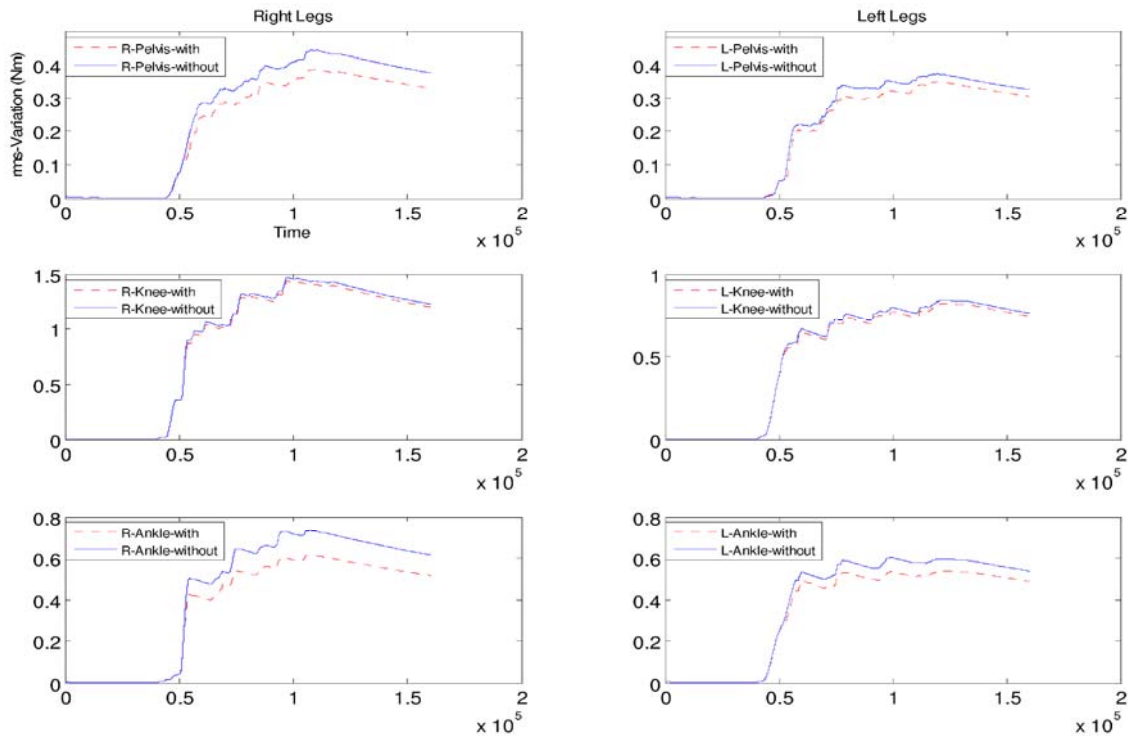


Fig. 4 RMS torque differences in the Pitch Joints : (red line: With-Constraint, blue line: Without-Constraint)

Table 2 Variation about RMS torque difference between With-Constraint and Without-Constraint equation

	With-Constraint	Without-Constraint	Variation
Right Pelvis	0.3304	0.3766	0.0462
Right Knee	1.2014	1.2282	0.0268
Right Ankle	0.5165	0.6151	0.0986
Left Pelvis	0.3059	0.3262	0.0203
Left Knee	0.7460	0.7640	0.018
Left Ankle	0.4881	0.5378	0.0497

6. Conclusion

In this paper, a new approach towards dynamic modeling of humanoid robots was presented, where the constraints are holonomic in the double support phase and due to the contact between the feet and surface of the ground. The presence of the constraint causes additional force of constraint to be applied to the equation of motion of the constrained system. Therefore, there are two components to analyze in the motion of humanoid robots: an unconstrained equation subjected to the external impressed force; and a constraint equation subject to the additional forces of constraints. They are composed of the fundamental equation of motion for designing humanoid robot dynamics. The central argument in the design of dynamic models is to find those constraints. The dynamic model of the humanoid robot is shown to be divided by the sagittal and frontal planes based on the fundamental equation in generalized coordinates. In practical situations, the dynamic model should be expected to deal with some trajectory tracking or optimization issues for stability under any external disturbance. The equation of motion obtained through the analytical method provides an explicit dynamic model of multi-body systems such as a humanoid robot. This needs to be extended to include non-ideal constraints, such as friction between links or between the feet and ground, to make the dynamic model more accurate. Unlike general control methods, this approach can be applied to trajectory tracking problems. Jan Peter³ contributed a derived control framework as a special case of optimal control based on the analytical approach. This method is based upon Udwadia's work.⁴ For future study, the next step is to apply the calculations in this paper to control the trajectory motion of humanoid robots.

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2. $m > n$, the solution is the one that minimizes the quantity, $\|b - AX\|$.
3. $m < n$, the solution is infinite number of solutions and MP solution is the particular solution whose vector 2-norm is minimal.

B.

The general solution to the consistent equation $AX = b$, where A is an m by n matrix.

$$X = A^+b + (I - A^+A)h,$$

where h is n by 1 vector

If $m \times n$ matrix A has rank n ,

$$A^+ = A^T(AA^T)^+ = (A^T A)^+ A^T$$

$$A^+ = (A^T A)^{-1} A^T$$

$$(A^T A)^{-1} = (AA^T)^+$$

$$AA^+ = (A^T A)^{-1} A^T A = I$$

$$\therefore I - AA^+ = 0$$

APPENDIX

A.

The Moore-Penrose Generalized Inverse is a general way to find the solution to the following system of linear equations

$$AX = b, \quad b \in \mathbb{R}^m; X \in \mathbb{R}^n; A \in \mathbb{R}^{m \times n}$$

If the matrix A^+ satisfies all the following four conditions, the MP inverse of A is the unique matrix.

$$1. AA^+A = A$$

$$2. A^+AA^+ = A^+$$

$$3. AA^+ = (AA^+)^T$$

$$4. A^+A = (A^+A)^T$$

If matrix A is $m \times n$,

1. $m = n$, $A^+ = A^{-1}$ if A is full rank